Toward a four-dimensional Yang–Mills construction

Notebook 04: strictly local Gaussian cutoff activities

Research working document

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Status: internally checked abstract theorem. This standalone notebook records a sufficient theorem and proof. The support algebra, inverse estimate, full Gaussian density majorant, forest expansion, and final support-root summation have received two substantive internal reviews, including an independent derivation/audit. This is not external mathematical certification. The hypotheses have not been matched to the actual Yang–Mills fluctuation operators, and no interacting cutoff-transfer theorem or continuum construction is claimed.

The purpose is to upgrade spatial localization in the positions of rare Gaussian cutoff defects to strict locality in the background variables. The extra hypothesis is a small sum of local perturbations of a fixed precision matrix. The proof expands the entire normalized marginal density, including its determinant and inverse. No derivative of a sharp indicator is taken.

1 Setting, constants, and conclusion

Let $\Lambda \subset \mathbb{Z}^d$ be finite. Distances are nearest-neighbor lattice graph distances, and each block carries D real coordinates, with D fixed. Every constant below is uniform in Λ. The background domain is a product

$$\mathcal{U}_\Lambda = \prod_{b\in\Lambda} \mathcal{U}_b$$

of open complex domains. A local function with support Y is a bounded holomorphic function of $A|_Y$ on $\prod_{b\in Y}\mathcal{U}_b$. Its uniform norm is denoted $\|\cdot\|_\infty$.

Let $C_{0,\Lambda} = Q_{0,\Lambda}^{-1}$ be background-independent, complex symmetric, and satisfy

$$\operatorname{Herm} C_{0,\Lambda} \geq \lambda I, \qquad \|C_{0,\Lambda}\| \leq \Lambda_0, \qquad 0 < \lambda \leq \Lambda_0 < \infty. \tag{1}$$

Real positive $C_{0,\Lambda}$ is the principal example. Fix $0 < \kappa < m$ such that the weighted block Schur bound

$$K := \sup_\Lambda \sup_{b\in\Lambda} \sum_{c\in\Lambda} e^{(\kappa+2m)d(b,c)} \left\|(C_{0,\Lambda})_{bc}\right\| < \infty \tag{2}$$

holds. Transpose symmetry gives the same column bound.

Consider the precision

$$Q_\Lambda(A) = Q_{0,\Lambda} + V_\Lambda(A), \qquad V_\Lambda(A) = \sum_{\varnothing\neq Y\subset\Lambda} V_Y(A|_Y), \tag{3}$$

where each V_Y is individually complex symmetric, is supported in $Y \times Y$ as a block matrix, and is holomorphic and local as above. Write $\ell(Y)$ for the minimum metric spanning-tree length of Y,

with $\ell(\{b\}) = 0$. Assume

$$\nu := \sup_{\Lambda} \sup_{b \in \Lambda} \sum_{Y \ni b} \sum_{c \in Y} e^{(\kappa+2m)\ell(Y)} \left\| (V_Y)_{bc} \right\|_\infty < \infty. \tag{4}$$

The same bound holds for columns. In particular the ordinary Schur norm of V_Λ is at most ν.

Here are explicit sufficient smallness conditions. Define

$$\delta = \lambda/4, \qquad v = \frac{K^2\nu}{1 - K\nu}, \qquad v_1 = v(1 + \delta/K), \tag{5}$$

$$s = \min\left\{ \frac{m}{2}, \frac{\lambda e m}{8(K+\delta)} \right\}, \qquad L = \frac{4}{\lambda} \sum_{z \in \mathbb{Z}^d} e^{-s|z|}, \qquad V_* = \frac{8\Lambda_0^2}{\lambda}. \tag{6}$$

Require

$$K\nu < 1, \qquad v_1 \le \lambda/4, \qquad u := Lv_1 \le \frac{1}{2}, \qquad w := \frac{L^2 v_1}{1-u} \le \frac{1}{2V_*}. \tag{7}$$

These conditions hold when ν is sufficiently small with the baseline constants fixed. Set

$$\widehat{\tau}_R = \left(\frac{16 V_*}{\lambda} \right)^{D/2} e^{-R^2/(8V_*)}, \tag{8}$$

$$A_\kappa = K \sum_{z \in \mathbb{Z}^d \setminus \{0\}} e^{-(m-\kappa)|z|}, \qquad B_0 = \frac{8e^2 A_\kappa}{\delta}, \qquad a_* = \min\{1/2, \kappa/4\}. \tag{9}$$

Finally require

$$\widehat{\tau}_R \le \frac{1}{2e^2 \max\{1, B_0\}}. \tag{10}$$

For a complex symmetric accretive covariance C, use the normalized Gaussian density

$$p_C(x) = (2\pi)^{-N/2} (\det C)^{-1/2} e^{-x^T C^{-1} x/2}. \tag{11}$$

The determinant branch is the analytic branch on the accretive cone that agrees with the positive square root on real positive matrices. This density integrates to one. Its real-domain integral is absolutely convergent because

$$\operatorname{Herm} C^{-1} = C^{-*} (\operatorname{Herm} C) C^{-1} > 0.$$

Put $C_\Lambda(A) = Q_\Lambda(A)^{-1}$ and define the fixed-domain cutoff partition function

$$Z_{\Lambda,R}(A) = \int_{\mathbb{R}^{D|\Lambda|}} p_{C_\Lambda(A)}(x) \prod_{b \in \Lambda} \mathbf{1}_{\{\|x_b\|_2 \le R\}} \, dx. \tag{12}$$

The real domain and threshold do not move with the complex background.

Theorem 1 (Proposed strict-locality theorem)**.** *Under (1)–(10), $Z_{\Lambda,R}$ has a holomorphic logarithm, fixed by turning on the defect strength from zero, and an absolutely convergent representation*

$$\log Z_{\Lambda,R}(A) = \sum_H E_{\Lambda,H}(A|_{H \cap \Lambda}). \tag{13}$$

Here H ranges over finite connected sets in $\mathbb{Z}^d$, and the activities are strictly local, bounded, and holomorphic in the indicated variable background. They satisfy

$$\sup_{b \in \mathbb{Z}^d} \sum_{H \ni b} e^{a_*|H|} \left\| E_{\Lambda,H} \right\|_\infty \le 2e^2 \left(\sum_{z \in \mathbb{Z}^d} e^{-(\kappa/2)|z|} \right) \widehat{\tau}_R. \tag{14}$$

One may take H in the lattice hull of Λ when Λ is not geodesically convex.

Remark 1 (Finite-volume dependence)**.** *The activity $E_{\Lambda,H}$ may depend on the entire* background-independent *baseline $C_{0,\Lambda}$. Strict locality asserts only that its* variable background dependence *reads $A|_{H\cap\Lambda}$. Neither equality of activities in different volumes nor existence of their thermodynamic limit follows from* (14)*. A consistency or comparison argument for the baselines is an additional obligation.*

Remark 2 (Product-domain hypothesis)**.** *The product background domain is an explicit sufficient hypothesis. A weak gauge-potential ball or a constrained gauge domain is not automatically such a product. An application must supply an appropriate product chart or polydomain, or separately verify the support algebra and local-function bounds on the actual domain.*

2 The support algebra and local covariance paths

For the proof, suppress Λ from the notation. Fix a nonempty marked set $S \subset \Lambda$. For a background support $X \subset \Lambda$, let $\ell_S(X)$ be the minimum total length of a metric forest on $S \cup X$ in which every component meets S. Roots in S may remain in different components, and $\ell_S(\varnothing) = 0$. We have

$$\ell_S(X \cup X') \le \ell_S(X) + \ell_S(X'), \qquad S \subset S' \Longrightarrow \ell_{S'}(X) \le \ell_S(X). \tag{15}$$

Consider formal families $F = (F_X)_X$ of bounded local holomorphic functions with norm

$$\|F\|_{\mathcal{B}_S} = \sum_X e^{\kappa \ell_S(X)} \|F_X\|_\infty . \tag{16}$$

Multiplication is union convolution, $(FG)_Z = \sum_{X\cup X'=Z} F_X G_{X'}$. Equation (15) makes the completion a commutative unital Banach algebra. Evaluation $F \mapsto \sum_X F_X(A|_X)$ is a bounded algebra homomorphism. Representations need not be unique; throughout use the explicit series in the V_Y, grouped by the union of their supports. Formal scalar variables multiplying the V_Y can be retained until absolute convergence has been established.

For matrices with entries in this support algebra, use the maximum of the row and column Schur norms, taking the block operator norm inside the support sum. This norm is submultiplicative.

The covariance is the convergent Neumann series

$$C(A) = \sum_{j\ge 0} (-1)^j C_0 (V C_0)^j . \tag{17}$$

Let $\Delta C = C - C_0$, and denote by ΔC_S its actual principal marginal on S.

Lemma 1 (Covariance paths with attached supports)**.** *The series* (17)*, with its explicit local supports, obeys*

$$\max\left\{ \sup_{b\in S} \sum_{c\in S} e^{2md(b,c)} \|(\Delta C_S)_{bc}\|_{\mathcal{B}_S} , \sup_{c\in S} \sum_{b\in S} e^{2md(b,c)} \|(\Delta C_S)_{bc}\|_{\mathcal{B}_S} \right\} \le v. \tag{18}$$

Proof. In a term $C_0 V_{Y_1} C_0 \cdots V_{Y_j} C_0$ with endpoints $b, c \in S$, choose a spanning tree for each Y_i. Together with the covariance links between consecutive perturbations these trees connect all background supports to the endpoints. If L_{path} is the sum of the covariance-link distances and $\sum_i \ell(Y_i)$, then

$$\ell_S\left(\bigcup_i Y_i\right) \le L_{\text{path}}, \qquad d(b,c) \le L_{\text{path}}.$$

Therefore

$$e^{\kappa \ell_S(\cup_i Y_i)+2md(b,c)} \le e^{(\kappa+2m)L_{\mathrm{path}}}.$$

Successive weighted matrix row sums using (2) and (4) bound the order-j contribution by $K^{j+1}\nu^j$. The column calculation is identical. Summing $j \ge 1$ gives $v = K^2\nu/(1-K\nu)$. □

In particular, $\|(\Delta C_S)_{bc}\|_{\mathcal{B}_S} \le ve^{-2md(b,c)}$. This pays for the support of the *full covariance marginal*. It is stronger than expanding only the covariance factors that will appear on a marked tree.

3 Tree interpolation and uniform inverse bounds

For a partition π of S, let v_π be the zero-one mask that retains covariance entries within each component and sets other entries to zero. The matrix $C_{0,S}(v_\pi)$ is a direct sum of principal submatrices, so it obeys (1). The same is true for every mask in the convex hull of the partition masks.

For a forest F and parameters $t_e \in [0,1]$, the BKAR mask is

$$s^F_{bc}(t) = \begin{cases} \min_{e\in \mathrm{path}_F(b,c)} t_e, & b,c \text{ in the same component,} \\ 0, & b,c \text{ in different components,} \end{cases} \tag{19}$$

with diagonal entries equal to one. It belongs to that convex hull: integrate, over $r \in [0,1]$, the partition mask of the subforest retaining edges with $t_e \ge r$. This gives (19). The forest formula may be applied directly on this partition polytope; no positivity assumption for every arbitrary coordinate-cube mask is needed [1, 2].

Fix a tree T on S, $n = |S| \ge 2$, with vertex degrees d_b. Put

$$J_{bc} = Ke^{-md(b,c)}, \qquad r_{\{b,c\}} = \frac{\delta}{\max(d_b,d_c)J_{bc}} \quad (\{b,c\} \in T). \tag{20}$$

At a BKAR point s^T, allow independent complex increments z_e on its tree edges with $|z_e| < r_e$. Let M_0 be the resulting masked baseline $C_{0,S}(s^T + z)$. Since

$$\sum_{c:\{b,c\}\in T} \frac{1}{\max(d_b,d_c)} \le 1,$$

the perturbation of $C_{0,S}(s^T)$ has row and column norms at most δ. Consequently

$$\operatorname{Herm} M_0 \ge \lambda I/2, \qquad \|M_0\| \le 2\Lambda_0. \tag{21}$$

Moreover, its Schur norm with entry weight $e^{md(b,c)}$ is at most $K+\delta$. For the perturbation part, this uses

$$\|(C_0)_{bc}\| \le Ke^{-(\kappa+2m)d(b,c)}$$

in (20).

Lemma 2 (Elementary weighted-conjugation inverse bound)**.** *Suppose* $\operatorname{Herm} M \ge aI$ *and the weighted Schur norm of* M *at rate* m *is at most* $\overline{K}$*. Let*

$$s_0 = \min\{m/2, aem/(4\overline{K})\}.$$

Then

$$\left\|(M^{-1})_{bc}\right\| \le (2/a)e^{-s_0 d(b,c)}. \tag{22}$$

Proof. Fix c and conjugate by the scalar block-diagonal weight $D_{bb} = e^{s_0 d(b,c)}$. The triangle inequality for the distance gives

$$\begin{aligned}\left\|DMD^{-1} - M\right\|_{\mathrm{Sch}} &\le \overline{K} \sup_{r\ge 0}(e^{s_0 r} - 1)e^{-mr} \\ &\le \frac{2s_0\overline{K}}{em} \le a/2.\end{aligned}$$

The middle inequality follows from $e^{s_0 r} - 1 \le s_0 r e^{s_0 r}$ and $\sup_{r\ge 0} r e^{-(m-s_0)r} \le 2/(em)$. As $\|M^{-1}\| \le 1/a$, a Neumann inverse yields $\|(DMD^{-1})^{-1}\| \le 2/a$. Its (b,c) block is $e^{s_0 d(b,c)}(M^{-1})_{bc}$, which proves (22). □

Apply Lemma 2 with $a = \lambda/2$ and $\overline{K} = K + \delta$. The resulting rate is the s in (6), and

$$\left\|M_0^{-1}\right\|_{\mathrm{Sch}} \le L. \tag{23}$$

This is uniform in S, T, t, z. The rate s may be much smaller than m or κ. Only the *ordinary* inverse Schur norm L is used below. All its matrix indices lie in the free root set S; its entries do not pay for attachment of background supports. Thus the proof does not require $\kappa < s$.

4 Expansion of the entire marginal Gaussian density

Apply the same mask $s^T + z$ to $C_S(A) = C_{0,S} + \Delta C_S$. Write the resulting matrix as $M = M_0 + E$. Lemma 1 and (20) imply

$$\|E\|_{\mathrm{Sch}(\mathcal{B}_S)} \le v_1 = v(1 + \delta/K). \tag{24}$$

The s^T part has norm at most v. The additional tree-edge row sums are bounded by $\delta v/K$, using the entry estimate $ve^{-2md(b,c)}$ and the degree sum above. The column bound is the same.

In the matrix support algebra set $F = M_0^{-1}E$. By (7), $\|F\|_{\mathrm{Sch}(\mathcal{B}_S)} \le u \le 1/2$, so the full inverse and determinant expansions converge:

$$M^{-1} - M_0^{-1} = \sum_{j\ge 1}(-F)^j M_0^{-1}, \qquad \left\|M^{-1} - M_0^{-1}\right\|_{\mathrm{Sch}(\mathcal{B}_S)} \le w, \tag{25}$$

$$\|\mathrm{Tr}\log(I+F)\|_{\mathcal{B}_S} \le Dn[-\log(1-u)]. \tag{26}$$

For (26), each trace is bounded by Dn times the matrix Schur norm, and $\sum_{j\ge 1} u^j/j = -\log(1-u)$.

The exact Gaussian density identity is

$$p_M(x) = p_{M_0}(x)\exp\left\{-\frac{1}{2}\mathrm{Tr}\log(I+F) - \frac{1}{2}x^T(M^{-1} - M_0^{-1})x\right\}. \tag{27}$$

It expands both the determinant and the inverse. It follows that

$$\|p_M(x)\|_{\mathcal{B}_S} \le |p_{M_0}(x)|\exp\left\{\frac{Dn}{2}[-\log(1-u)] + \frac{w}{2}\|x\|^2\right\}. \tag{28}$$

The quadratic estimate uses the row-and-column Schur bound in (25), applied to the block Euclidean norms of x.

Equation (21) gives

$$\mathrm{Herm}\, M_0^{-1} = M_0^{-*}(\mathrm{Herm}\, M_0)M_0^{-1} \ge V_*^{-1} I, \qquad |\det M_0| \ge (\lambda/2)^{Dn}.$$

Using $u \le 1/2$ and $w \le 1/(2V_*)$, the right-hand side of (28) is bounded by $(8V_*/\lambda)^{Dn/2}$ times the independent real Gaussian density with covariance $2V_*I$. For one D-dimensional block G of that real Gaussian,

$$\mathbb{P}(\|G\| > R) \le e^{-R^2/(8V_*)}\mathbb{E}e^{\|G\|^2/(8V_*)} = 2^{D/2}e^{-R^2/(8V_*)}. \tag{29}$$

Set $q_b(x_b) = -\mathbf{1}_{\{\|x_b\|>R\}}$. Combining (28) and (29) proves the support-valued all-marked bound

$$\left\| \int p_M(x) \prod_{b\in S} q_b(x_b)\, dx \right\|_{\mathcal{B}_S} \le \widehat{\tau}_R^{\,n}. \tag{30}$$

All coefficient sums, integrals, and matrix series can be interchanged using these absolute majorants. The indicators remain fixed measurable multipliers. The same bound covers $n = 1$, with no tree increments.

The inequality $v_1 \le \lambda/4$ also ensures that evaluation of these matrices retains accretivity on the required interpolation domains. The determinant branches in (27) agree with the normalized Gaussian branch, by continuation from zero perturbation.

5 Exact marked polymers and tree estimates

For a nonempty S, define

$$f_S(s;A) = \int p_{C_S(s;A)}(x) \prod_{b\in S} q_b(x_b)\, dx, \qquad m(S) = f_S(1;A).$$

These functions are holomorphic in a neighborhood of the partition polytope. The BKAR formula [1, 2] gives a sum over forests. At a forest mask the covariance matrix is block diagonal along forest components. Derivatives in the forest edges remain internal to those components. Therefore the forest sum factorizes exactly as

$$m(S) = \sum_{\pi \text{ partition of } S} \prod_{B\in\pi} \zeta(B), \tag{31}$$

where $\zeta(\{b\}) = m(\{b\})$ and

$$\zeta(B) = \sum_{T \text{ tree on } B} \int_{[0,1]^T} \partial_T f_B(s^T(t))\, dt \qquad (|B| \ge 2). \tag{32}$$

This identity also holds coefficientwise in the formal source series. Although covariance blocks can read overlapping background supports, the masked Gaussian matrix itself is block diagonal.

Banach-valued Cauchy applied to (30), in the polydisc (20), yields

$$\left\|\partial_T f_S(s^T)\right\|_{\mathcal{B}_S} \le \widehat{\tau}_R^{\,n}\delta^{-(n-1)} \prod_{\{b,c\}\in T} \max(d_b, d_c) J_{bc}.$$

Each parameter is differentiated once, so no factorial appears from Cauchy. For integer $d \ge 1$, $d^d \le e^d d!$, while $\sum_b d_b = 2n-2$ and $\max(d_b,d_c) \le d_b d_c$. Thus

$$\left\|\partial_T f_S(s^T)\right\|_{\mathcal{B}_S} \le \widehat{\tau}_R^{\,n}(e^2/\delta)^{n-1} \prod_{b\in S} d_b! \prod_{e\in T} J_e. \tag{33}$$

For completeness, the exact degree-weighted tree count is

$$\sum_{T \text{ tree on } [n]} \prod_{i=1}^{n} d_i! = (n-2)! \binom{3n-3}{n-2}. \tag{34}$$

Indeed write $d_i = k_i + 1$ with $\sum_i k_i = n-2$. The Pruefer count for this degree sequence is $(n-2)!/\prod_i k_i!$. Multiplication by $\prod_i d_i!$ leaves $(n-2)! \prod_i (k_i+1)$, whose sum is the coefficient of z^{n-2} in $(1-z)^{-2n}$. In particular,

$$\frac{1}{(n-1)!} \sum_T \prod_i d_i! = \frac{1}{n-1} \binom{3n-3}{n-2} \le 8^{n-1}. \tag{35}$$

To sum marked sets of size n containing a fixed root, label the other $n-1$ vertices and divide by $(n-1)!$. Dropping distinctness only increases the upper bound; set the edge majorant at coincident labels to zero. For each fixed tree, successive leaf sums of $e^{\kappa d(b,c)} J_{bc}$ are bounded by A_κ^{n-1}. Equations (33)–(35) imply

$$\sup_b \sum_{\substack{S \ni b \\ |S|=n}} e^{\kappa \ell(S)} \, \|\zeta(S)\|_{\mathcal{B}_S} \le \widehat{\tau}_R^{\,n} B_0^{n-1}. \tag{36}$$

The singleton bound is (30). Writing the canonical support representation as $\zeta(S) = \sum_X \zeta(S;X)(A|_X)$, we obtain

$$\sup_b \sum_{S \ni b, X} e^{\kappa \ell(S) + \kappa \ell_S(X) + \alpha |S|} \, \|\zeta(S;X)\|_\infty \le \frac{\widehat{\tau}_R e^\alpha}{1 - B_0 \widehat{\tau}_R e^\alpha} \tag{37}$$

whenever the denominator is positive.

Expanding $\prod_b (1+q_b)$ and using (31) gives the exact hard-core gas

$$Z_{\Lambda,R} = \sum_{\{S_1,\dots,S_k\} \text{ pairwise disjoint}} \prod_i \zeta(S_i). \tag{38}$$

The empty family has weight one. Compatibility concerns the *marked sets* S_i, not the enlarged background supports. Replacing it by disjointness of the enlarged supports would change the partition function.

6 The logarithm and background-rooted bounds

Use the scalar majorants

$$\widetilde{z}(S) = e^{\kappa \ell(S) + |S|} \sum_X e^{\kappa \ell_S(X)} \, \|\zeta(S;X)\|_\infty, \qquad a(S) = |S|.$$

Equation (37) with $\alpha = 2$ and (10) imply

$$\sum_{S' : S' \cap S \neq \varnothing} \widetilde{z}(S') e^{|S'|} \le |S| \, 2e^2 \widehat{\tau}_R \le |S|. \tag{39}$$

The hard-core polymer criterion and rooted cluster estimate [3] give the absolutely convergent Mayer logarithm

$$\log Z_{\Lambda,R} = \sum_{k \ge 1} \frac{1}{k!} \sum_{S_1,\dots,S_k} \varphi^T(S_1,\dots,S_k) \prod_i \zeta(S_i). \tag{40}$$

Here φ^T is the sum over connected graphs on $[k]$ of the product of $-\mathbf{1}_{\{S_i\cap S_j\neq\varnothing\}}$ over edges. Repeated marked polymers are allowed and are incompatible with themselves. The convergent expansion is the logarithm obtained by turning on an auxiliary defect strength from zero; it also proves zero-freeness on that defect polydisc.

Expand each $\zeta(S_i)$ in its support series and group by

$$M=\bigcup_i S_i, \qquad X=\bigcup_i X_i,$$

obtaining $U(M;X)(A|_X)$. The set M is always nonempty. For a connected overlap graph of the marked polymers,

$$\ell(M)\le\sum_i \ell(S_i), \qquad \ell_M(X)\le\sum_i \ell_{S_i}(X_i), \qquad |M|\le\sum_i |S_i|. \tag{41}$$

For the first inequality, combine the trees on the S_i; overlapping components need no extra connecting edges. For the second, a forest attached to S_i is also attached to the larger root set M, and (15) applies. The scalar majorant and rooted cluster bound consequently imply

$$\sup_b \sum_{M\ni b,X} e^{\kappa[\ell(M)+\ell_M(X)]+|M|}\,\|U(M;X)\|_\infty \le 2e^2\widehat{\tau}_R. \tag{42}$$

To check the rooting, a cluster whose marked union contains b has at least one polymer containing b. Choosing one such polymer overcounts. Its rooted Mayer sum with coefficient $1/(k-1)!$ is at most $e^{a(S)}$. Summing its weighted activity uses exactly (37) at $\alpha=2$, so no uncontrolled factor in the number of marks or polymers is introduced.

Choose deterministic minimizing metric trees on M and minimizing forests attaching X to M, and realize their edges by deterministic shortest lattice paths. Their union $H(M,X)$ is connected and contains $M\cup X$. If

$$L(M,X)=\ell(M)+\ell_M(X),$$

then

$$|H(M,X)|\le|M|+L(M,X), \qquad d(a,b)\le L(M,X) \quad (a\in M,\ b\in H(M,X)). \tag{43}$$

If paths leave a nonconvex Λ, the coefficient still reads only the original support $X\subset\Lambda$.

Group all $U(M;X)$ with the same $H(M,X)$ into $E_{\Lambda,H}$. This is a strictly local holomorphic function of $A|_{H\cap\Lambda}$. For $a\in M$, $b\in H$, and $a_*=\min\{1/2,\kappa/4\}$,

$$e^{a_*|H|}\le e^{|M|+\kappa L(M,X)}e^{-(\kappa/2)d(a,b)}. \tag{44}$$

Choose one marked a, or overcount by all $a\in M$. Sum first over the clusters containing this a using (42), and then over a. Equation (44) proves (14). This pays explicitly for a root lying in an added background support or connecting path rather than in a marked block.

All series converge locally uniformly in the background and absolutely in the displayed majorants. They therefore preserve holomorphy and justify grouping and evaluation. This completes the proof of Theorem 1.

7 Why no disconnected unmarked component remains

The density expanded in (27) is the normalized *marginal on a nonempty marked set*. Every Neumann path in ΔC_S starts and ends in S. Every background support in its coefficients is thus attached to

S before the inverse and determinant are expanded. Multiplication in $\mathcal{B}_S$ preserves this attachment property.

The factors M_0^{-1} and the trace in the determinant series operate only on the already designated roots S. They cannot create a component consisting solely of remote background supports. In the subsequent logarithm every gas polymer contains at least one mark. There is no $M = \varnothing$ term. This is the precise cancellation supplied by normalized marginals; one need not begin with a bulk determinant and cancel disconnected source-only loops afterward.

8 Limits of the theorem and the model interface

The theorem is a sufficient statement for a Gaussian cutoff normalizer. Notebooks 06 and 07 now provide its positive, bounded, exponentially local homogeneous flat reference on the specified retained coordinates, including a free refinement limit at rate $O(M^{-2})$ in weighted kernel norm. The physical identification has the smooth-minimizer and coordinate requirements stated there. The following points remain separate in a Yang–Mills application.

1. The actual independent-coordinate precision must admit (3) and the uniformly small norm (4) on the required complex domain. A spectral gap and covariance decay alone do not imply that local perturbation structure.

2. Gauge constraints and coordinate choices must lead to fixed block dimension and a fixed real cutoff domain during complex background continuation. A moving cutoff domain is not covered.

3. Baseline and perturbation constants must be uniform in scale, history, and volume, and fixed before the target remainder tolerance.

4. Nonquadratic interaction polymers require a joint expansion and stability argument. This notebook proves no such interacting transfer.

5. The positive exponent a_* in (14) must be compared with the numerical polymer norm actually needed downstream. A stronger target exponent requires enough initial decay.

6. Strict background locality does not imply consistency between different finite volumes. The baseline $C_{0,\Lambda}$ can change, affecting $E_{\Lambda,H}$ even for fixed H. Likewise, if a baseline is allowed to depend on an external real reference configuration, locality in the perturbation variables does not establish locality in that reference configuration. Applying the theorem around each reference separately cannot remove this additional dependence.

7. A weak gauge-potential ball or constrained current domain is not silently identified with the product domain used here. An appropriate chart or a separate support-algebra argument is needed.

8. The theorem does not assert activity-level gauge or lattice rotation invariance. Deterministic connecting paths can preserve translations while breaking rotations. Any equivariant construction or averaging step must preserve the actual domain, support labels, and weighted norm. Gauge invariance of the total Gaussian functional alone does not establish gauge invariance of each activity.

Covariance entry decay alone cannot replace the locality hypothesis. For example,

$$C_{bc}(A) = (1 + \varepsilon A_{x_*}) I_D \, \mathbf{1}_{\{b=c\}}, \qquad |A_{x_*}| < 1, \quad 0 < \varepsilon < 1/2,$$

is uniformly accretive, bounded, holomorphic, and has zero off-diagonal covariance. Yet every singleton cutoff logarithm depends on the arbitrarily remote background coordinate A_{x_*}. The local precision hypothesis excludes precisely this type of uncontrolled remote dependence.

Finally, if $R(g) = c_0(\log g^{-2})^p$ with $p > 1/2$ and $c_0 > 0$, then the right-hand side of (14) is a fixed constant times $\exp[-c_0^2(\log g^{-2})^{2p}/(8V_*)]$. It is smaller than any prescribed fixed multiple of $g(\log g^{-2})^q$ for sufficiently small g, for every fixed real q. The threshold is independent of a remainder tolerance only when all preceding constants were chosen independently of that tolerance.

References

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